

Penalty view

$$C_k \rightarrow \lambda_k \rightarrow \Sigma$$

$$\lambda_k = \beta_k \quad (c = 0)$$

exact settings

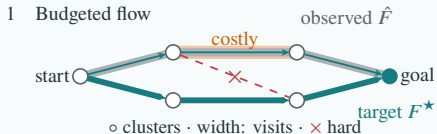
$\beta_k = \text{const}$

Fixed penalty

adaptive β_k

Lagrangian

SafeOT-Dual: one price rule



2 price

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

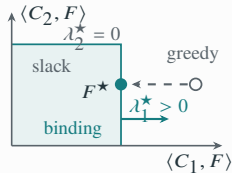
clipped c , filtered

feedback $\bar{J}_k - B_k$

3

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A c_k}{1 + \sum_k \lambda_k} \rightarrow \text{PPO/TRPO}$$

Constraint view



$\varepsilon \rightarrow 0$: LP dual

clip = c : active set

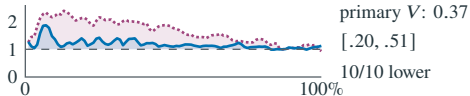
conditional limits

Training cost/ B

— SafeOT-Dual TRPO-Lag ($c = 0$)

shaded: median excess

SafePO (10 tasks)



Bullet (6 tasks)

